

5-11, 12, 14, 15, 16

$$11) T(s) = \frac{G(s)}{1 + G(s)H(s)} \quad G(s) = \frac{225}{s(s+15)}$$

$$T(s) = \frac{\frac{225}{s(s+15)}}{\frac{225 + s^2 + 15s}{s(s+15)}}$$

$$\omega_n = \sqrt{225} = 15 \quad 2\zeta\omega_n = 15 \Rightarrow \zeta = \frac{1}{2}$$

$$\%OS = \exp\left(-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right) \cdot 100$$

$$\%OS = \exp\left(-\frac{0.5 \cdot \pi}{\sqrt{1-0.25}}\right) \cdot 100 = 16.303\%$$

$$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{15/2} = \frac{8}{15} \text{ sec}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1-\zeta^2}} = \frac{\pi}{15\sqrt{1-0.25}} = 0.242 \text{ sec}$$

The system is underdamped. The transfer function has poles with real and imaginary parts.

$$12) T(s) = \frac{G(s)}{1 + G(s)H(s)}$$

$$T(s) = \frac{\frac{4}{s(s+3)}}{1 + \frac{4}{s(s+3)}} = \frac{4}{s^2 + 3s + 4}$$

$$C(s) = \frac{1}{s} \cdot \frac{4}{s^2 + 3s + 4} = \frac{4}{s^3 + 3s^2 + 4s} = \frac{4}{s(s^2 + 3s + 4)}$$

$$s = \frac{-3 \pm \sqrt{9 - 4 \cdot 1 \cdot 4}}{2} = -\frac{3}{2} \pm \frac{\sqrt{5}}{2}i$$

$$C(s) = \frac{4}{s(s - (-\frac{3}{2} + \frac{\sqrt{5}}{2}i))(s - (-\frac{3}{2} - \frac{\sqrt{5}}{2}i))}$$

$$\text{Res}(p_1) = \lim_{s \rightarrow 0} \frac{4}{s^2 + 3s + 4} \cdot e^{st} = 1$$

$$\text{Res}(p_2) = \lim_{s \rightarrow -\frac{3}{2} + \frac{\sqrt{5}}{2}i} \frac{4}{s(s + \frac{3}{2} + \frac{\sqrt{5}}{2}i)} \cdot e^{st} =$$

$$\frac{4}{-\frac{3}{2} + \frac{\sqrt{5}}{2}i(\sqrt{5}i)} e^{(-\frac{3}{2} + \frac{\sqrt{5}}{2}i)t} = \frac{4}{-\frac{3}{2} + \frac{\sqrt{5}}{2}i} e^{(-\frac{3}{2} + \frac{\sqrt{5}}{2}i)t}$$

$$= \frac{8}{-5 - 3\sqrt{5}i} e^{-\frac{3}{2}t} \left(\cos\left(\frac{\sqrt{5}}{2}t\right) + i \sin\left(\frac{\sqrt{5}}{2}t\right) \right)$$

$$\text{Res}(p_3) = \lim_{s \rightarrow -\frac{3}{2} - \frac{\sqrt{5}}{2}i} \frac{4}{s(s - (-\frac{3}{2} + \frac{\sqrt{5}}{2}i))} e^{st} = \frac{4}{-\frac{3}{2} - \frac{\sqrt{5}}{2}i(-\sqrt{5}i)} e^{(-\frac{3}{2} - \frac{\sqrt{5}}{2}i)t}$$

$$\frac{4}{\frac{3}{2}i - \frac{5}{2}} e^{(-\frac{3}{2} - \frac{\sqrt{5}}{2}i)t} = \frac{8}{-5 + 3\sqrt{5}i} e^{-\frac{3}{2}t} \left(\cos\left(\frac{\sqrt{5}}{2}t\right) + i \sin\left(\frac{\sqrt{5}}{2}t\right) \right)$$

$$C(t) = 1 + e^{-\frac{3}{2}t} \left[\frac{8}{-5 - 3\sqrt{5}i} \left(\cos\left(\frac{\sqrt{5}}{2}t\right) + i \sin\left(\frac{\sqrt{5}}{2}t\right) \right) + \frac{8}{-5 + 3\sqrt{5}i} \left(\cos\left(\frac{\sqrt{5}}{2}t\right) + i \sin\left(\frac{\sqrt{5}}{2}t\right) \right) \right]$$

$$C(t) = 1 + e^{-\frac{3}{2}t} \left[\frac{8}{-5 - 3\sqrt{5}i} \cos\left(\frac{\sqrt{5}}{2}t\right) + i \frac{8}{-5 - 3\sqrt{5}i} \sin\left(\frac{\sqrt{5}}{2}t\right) + \frac{8}{-5 + 3\sqrt{5}i} \cos\left(\frac{\sqrt{5}}{2}t\right) + i \frac{8}{-5 + 3\sqrt{5}i} \sin\left(\frac{\sqrt{5}}{2}t\right) \right]$$

$$c(t) = 1 + e^{-\frac{3}{2}t} \left[\left(\frac{8}{-s-3\sqrt{5}i} + \frac{8}{-s+3\sqrt{5}i} \right) \cos\left(\frac{\sqrt{5}}{2}t\right) + i \left(\frac{8}{-s-3\sqrt{5}i} - \frac{8}{-s+3\sqrt{5}i} \right) \sin\left(\frac{\sqrt{5}}{2}t\right) \right]$$

$$\frac{8}{-s-3\sqrt{5}i} + \frac{8}{-s+3\sqrt{5}i}$$

$$\frac{8(-s+3\sqrt{5}i) + 8(-s-3\sqrt{5}i)}{250} = \frac{-80}{250}$$

$$c(t) = 1 + e^{-\frac{3}{2}t} \left[-\frac{80}{250} \cos\left(\frac{\sqrt{5}}{2}t\right) - \frac{80}{250} i \sin\left(\frac{\sqrt{5}}{2}t\right) \right]$$

$$c(t) = 1 - e^{-\frac{3}{2}t} \left[\frac{80}{250} \cos\left(\frac{\sqrt{5}}{2}t\right) + \frac{80}{250} i \sin\left(\frac{\sqrt{5}}{2}t\right) \right]$$

$$14) T(s) = \frac{K}{s(s+30)} = \frac{K}{s^2+30s+K} \quad \begin{aligned} \omega_n &= \sqrt{K} \\ 2\zeta\omega_n &= 30 \\ \zeta &= \frac{15}{\sqrt{K}} \end{aligned}$$

$$\%OS = \exp\left(-\frac{\zeta\pi}{\sqrt{1-\zeta^2}}\right) \times 100 =$$

$$0.1 = \exp\left(-\frac{\frac{15}{\sqrt{K}} \cdot \pi}{\sqrt{1 - \left(\frac{15}{\sqrt{K}}\right)^2}}\right)$$

$$0.1 = \exp\left(-\frac{15\pi/\sqrt{K}}{\sqrt{1 - 225/K}}\right)$$

$$\ln(0.1) = -\frac{15\pi/\sqrt{K}}{\sqrt{1 - 225/K}}$$

$$\ln(0.1)\sqrt{1 - 225/K} = -\frac{15\pi}{\sqrt{K}}$$

$$\ln(0.1) \cdot \sqrt{K - 225} = -15\pi$$

$$K - 225 = \left(-\frac{15\pi}{\ln(0.1)}\right)^2$$

$$K = \left(-\frac{15\pi}{\ln(0.1)}\right)^2 + 225$$

$$K = 643.84$$

$$15) T(s) = \frac{K}{s(s+\alpha)} = \frac{K}{s^2+\alpha s+K} \quad \begin{aligned} \omega_n &= \sqrt{K} \\ 2\zeta\omega_n &= \alpha \\ \zeta &= \frac{\alpha}{2\sqrt{K}} \end{aligned}$$

$$T_s = \frac{4}{\zeta\omega_n} = \frac{4}{\frac{\alpha}{2\sqrt{K}} \cdot \sqrt{K}} = \frac{4}{\frac{\alpha}{2}} = \frac{8}{\alpha} \quad 0.12 = \frac{8}{\alpha} \Rightarrow \frac{8}{0.12} = \alpha$$

$$\alpha = \frac{200}{3}$$

$$\ln(0.2) = -\frac{\zeta\pi}{\sqrt{1-\zeta^2}} \Rightarrow \ln(0.2) = -\frac{\frac{100}{3\sqrt{K}} \cdot \pi}{\sqrt{1 - \frac{10000}{9K}}}$$

$$K = \left(-\frac{100\pi}{3\ln(0.2)}\right)^2 + \frac{10000}{9}$$

$$K = 5344.7 \quad \alpha = \frac{200}{3}$$

$$16) T(s) = K_1 \cdot \frac{50}{s(s+1)} = \frac{K_1 50}{s^2 + (1+50K_2)s} \quad \begin{aligned} \omega_n &= \sqrt{K_1 50} \\ 2\omega_n \zeta &= (1+50K_2) \\ \zeta &= \frac{(1+50K_2)}{2\sqrt{K_1 50}} \end{aligned}$$

$$1(s) = K_1 \cdot \frac{1}{1 + K_2 s} = \frac{1}{s^2 + (1 + 50K_2)s} \quad 2\omega_n \zeta = (1 + 50K_2) \quad \zeta = \frac{(1 + 50K_2)}{2\sqrt{K_1 \cdot 50}}$$

$$T_p = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{\pi}{\sqrt{K_1 \cdot 50} \sqrt{1 - \left(\frac{1 + 50K_2}{2\sqrt{K_1 \cdot 50}}\right)^2}}$$

$$T_p = \frac{\pi}{\sqrt{K_1 \cdot 50 - \frac{1 + 100K_2 + 2500K_2^2}{4}}}$$

$$K_1 \cdot 50 - \frac{1 + 100K_2 + 2500K_2^2}{4} = \pi^2$$

$$K_1 \cdot 50 - \frac{1}{4} - 25K_2 - 625K_2^2 = \pi^2$$

$$625K_2^2 + 25K_2 + \frac{1}{4} - K_1 \cdot 50 - \pi^2 = 0$$

$$K_2 = \frac{-25 \pm \sqrt{625 - 4 \cdot 625 \cdot \left(\frac{1}{4} - K_1 \cdot 50 - \pi^2\right)}}{2 \cdot 625}$$

$$K_2 = -\frac{1}{50} \pm \frac{\sqrt{1 + K_1 \cdot 200 - 4\pi^2}}{50}$$

$$K_2 = \frac{1}{50} \left[1 \pm \sqrt{200K_1 - 4\pi^2} \right]$$

$$K_2 = \frac{1}{50} \left[-1 + 2\sqrt{50K_1 - \pi^2} \right]$$

$$T_s = \frac{4}{3\omega_n} = \frac{4}{\sqrt{50K_1} (1 + 50K_2)} = \frac{8}{1 + 50K_2}$$

$$K_2 = \frac{3}{50}$$

$$2 = \sqrt{50K_1 - \pi^2} \Rightarrow 4 = 50K_1 - \pi^2 \quad K_1 = \frac{4 - \pi^2}{50} = 0.117$$

$$18) T(s) = \frac{7350}{s^2 + 100s + 7350} \quad \omega_n = \sqrt{7350} \quad 2\zeta\omega_n = 100 \quad \zeta = \frac{50}{\sqrt{7350}} \\ \omega_n = 85.7 \quad \zeta = 0.58$$

$$\%OS = \exp\left(-\frac{0.58 \cdot \pi}{\sqrt{1 - 0.58^2}}\right) \times 100 = 10.48\%$$

$$T_p = \frac{\pi}{85.7 \sqrt{1 - 0.58^2}} = 0.045 \text{ sec}$$

$$T_r = \frac{\pi - \tan^{-1}\left(\frac{\sqrt{1 - 0.58^2}}{0.58}\right)}{85.7 \sqrt{1 - 0.58^2}} = 0.0315 \text{ sec}$$

$$T_s = \frac{4}{\omega_n \zeta} = \frac{4}{85.7 \cdot 0.58} = 0.08 \text{ sec}$$